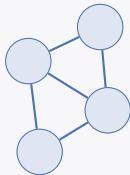


Input representation

Traffic scene graph $G = (V, E)$



Node features $\mathbf{x}_v \in \mathbb{R}^7$

speed | actor type (4) | lane-change | on-intersection

Edge features $\mathbf{e}_{uv} \in \mathbb{R}^7$

path length | edge type (6, one-hot)

Shared GINE encoder f_θ (weights tied across views)

Stochastic augmentation

View $\tilde{G}^a$

Gaussian noise $\sigma=0.08$

View $\tilde{G}^b$

edge drop $p=0.1$

GINE layer $\ell=1, \dots, 5$ ($d=384$)

$$\mathbf{x}_i^{(\ell)} = \text{MLP}^{(\ell)} \left[(1+\epsilon) \mathbf{x}_i^{(\ell-1)} + \sum_{j \in \mathcal{N}(i)} \text{ReLU}(\mathbf{x}_j^{(\ell-1)} + W_e \mathbf{e}_{ji}) \right]$$

Lin $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ Lin

$\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ Dropout_{0.2}

$\times 5$ layers

Graph readout

mean_v

max_v

sum_v

$\mathbf{h}_G \in \mathbb{R}^{3d=1152}$

Embedding MLP

Lin 1152 $\rightarrow$ 192 $\rightarrow$ ReLU
 $\rightarrow$ Drop_{0.2} $\rightarrow$ Lin 192 $\rightarrow$ 192

ℓ_2

Embedding z

ℓ_2 -norm, $\mathbb{R}^{192}$

used for coverage analysis / retrieval

Projection head g_ψ

Lin 192 $\rightarrow$ 192 $\rightarrow$ ReLU
 $\rightarrow$ Lin 192 $\rightarrow$ 192

projections $\mathbf{p}^a, \mathbf{p}^b$ of the two views

Contrastive loss (NT-Xent / InfoNCE, $\tau=0.07$)

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(\mathbf{p}^a, \mathbf{p}^b)/\tau)}{\sum_k \exp(\text{sim}(\mathbf{p}^a, \mathbf{p}^k)/\tau)}$$

Tensor

Message passing

Pooling

MLP head

Training loss

Augmentation